

Additional file 1: Supplementary tables

No Single Model Family Dominates: Ensembles and Conventional Machine Learning Perform Comparably to Pretrained Molecular Models Across 44 Property-Prediction Benchmarks

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This file contains the eight supplementary tables referenced in the main text. Tables S1–S7 are generated directly from the deposited benchmark artifacts by `portable_colab_qsar_bundle/render_manuscript_assets.py` and `render_latex_tables.py`; Table S8 is generated by `python -m qsarena.reliability_tables` from the reliability-study artifacts. They carry the same values as the machine-readable CSV and Markdown files in the repository.

All results derive from a single split and a single seed per dataset; as set out in the Limitations section of the main text, individual entries are therefore provisional and close comparisons between models should not be treated as established.

Table S1: Per-dataset winning model under the ranking metric, with the corresponding cross-validation-selected model and its relative gap to the dataset best (%).

dataset	suite	task - kind	family	model	analysis - analysis_metric_value	cv - selected - model	cv - selected - family	cv_selected_gap_to_best
tdc_ames	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_roc_auc	Extra trees	Conventional ML	0.577
tdc_bbb_martins	TDC	classification	Conventional ML	Random forest	test_roc_auc	Tabular MLP	Conventional ML	2.943
tdc_bioavailability_ma	TDC	classification	Conventional ML	CatBoost	test_roc_auc	Logistic Regression	Conventional ML	7.943
tdc_carcinogens_lagunin	TDC	classification	Chemprop v2 GNN	Chemprop v2 (AttentiveFP, ensemble=3)	test_roc_auc	Voting Classifier (KNN, SVM)	Conventional ML	9.918
tdc_clintox	TDC	classification	CFA combinatorial fusion	CFA (Combinatorial Fusion)	test_roc_auc	Logistic Regression	Conventional ML	8.005
tdc_cyp1a2_veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_roc_auc	ChemML MLP (Tensor Flow)	Deep tabular NN (ChemML MLP)	1.432
tdc_cyp2c19_veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_roc_auc	XGBoost	Conventional ML	8.976
tdc_cyp2c9_substrate_carbonmangels	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_auprc	SVC	Conventional ML	35.513
tdc_cyp2c9_veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_auprc	AdaBoost	Conventional ML	13.326
tdc_cyp2d6_substrate_carbonmangels	TDC	classification	Uni-Mol (3D pretrained)	Uni-Mol V1	test_auprc	AdaBoost	Conventional ML	4.966
tdc_cyp2d6_veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_auprc	AdaBoost	Conventional ML	16.154
tdc_cyp3a4_substrate_carbonmangels	TDC	classification	Conventional ML	Random forest	test_auprc	Logistic Regression	Conventional ML	2.791
tdc_cyp3a4_veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_auprc	AdaBoost	Conventional ML	9.750

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dataset	suite	task - kind	family	model	analysis - metric	analysis - metric - value	cv - selected - model	cv - selected - family	cv_selected_gap_to_best
tdc_dili	TDC	classification	Uni-Mol (3D pretrained)	Uni-Mol V1	test_roc_auc	0.922	Logistic Regression	Conventional ML	19.151
tdc_herg	TDC	classification	Conventional ML	AdaBoost	test_roc_auc	0.848	Logistic Regression	Conventional ML	16.386
tdc_herg_karim	TDC	classification	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_roc_auc	0.905	ChemML MLP (Py Torch)	Deep tabular NN (ChemML MLP)	3.672
tdc_hia_hou	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_roc_auc	0.988	Logistic Regression	Conventional ML	0.500
tdc_pampa_ncats	TDC	classification	Uni-Mol (3D pretrained)	Uni-Mol V1	test_roc_auc	0.763	Logistic Regression	Conventional ML	26.281
tdc_pgp_broccatelli	TDC	classification	Uni-Mol (3D pretrained)	Uni-Mol V2 (84m)	test_roc_auc	0.933	Logistic Regression	Conventional ML	6.966
tdc_skin_reaction	TDC	classification	Uni-Mol (3D pretrained)	Uni-Mol V2 (84m)	test_roc_auc	0.658	Voting Classifier (KNN, SVM)	Conventional ML	23.497
ω tdc_tox21	TDC	classification	Conventional ML	AdaBoost	test_roc_auc	0.556	SVC	Conventional ML	15.985
tdc_toxcast	TDC	classification	Conventional ML	HistGradientBoosting	test_roc_auc	0.716	Logistic Regression	Conventional ML	16.900
chemml_cep_homo	ChemML	regression	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_rmse	0.099	ElasticNetCV	Conventional ML	22.270
chemml_organic_density	ChemML	regression	Chemprop v2 GNN	Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)	test_rmse	0.005	ElasticNetCV	Conventional ML	34.525
esol_delaney	MoleculeNet	regression	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_rmse	0.617	ElasticNetCV	Conventional ML	17.596
freesolv_sampl	MoleculeNet	regression	Deep tabular NN (ChemML MLP)	ChemML MLP (Py Torch)	test_rmse	0.993	CatBoost	Conventional ML	115.432
lipophilicity	MoleculeNet	regression	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_rmse	0.589	ElasticNetCV	Conventional ML	15.769
poduam_pod_nc_std	PODUAM	regression	Conventional ML	Random forest	test_rmse	0.720	ElasticNetCV	Conventional ML	9.929
poduam_pod_rd_std	PODUAM	regression	Conventional ML	Random forest	test_rmse	0.572	ElasticNetCV	Conventional ML	22.906

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dataset	suite	task_ kind	family	model	analysis_ metric	analysis_ metric_ value	cv_ selected_ model	cv_ selected_ family	cv_selected_gap_to_best
polaris_adme_ fang_hppb_1	Polaris	regression	MapLight + GNN	MapLight + GNN (Cat Boost, Strict Parity)	test_ rmse	0.448	ElasticNetCV	Conventional ML	22.177
polaris_adme_ fang_perm_1	Polaris	regression	Uni-Mol (3D pretrained)	Uni-Mol V1	test_ rmse	0.399	ChemML MLP (Py Torch)	Deep tabular NN (ChemML MLP)	15.413
polaris_adme_ fang_rclint_1	Polaris	regression	Uni-Mol (3D pretrained)	Uni-Mol V1	test_ rmse	0.522	ElasticNetCV	Conventional ML	15.664
polaris_adme_ fang_rppb_1	Polaris	regression	MapLight + GNN	MapLight + GNN (Cat Boost, Strict Parity)	test_ rmse	0.493	ElasticNetCV	Conventional ML	21.893
polaris_adme_ fang_solu_1	Polaris	regression	Uni-Mol (3D pretrained)	Uni-Mol V2 (84m)	test_ rmse	0.551	ChemML MLP (Py Torch)	Deep tabular NN (ChemML MLP)	16.215
tdc_caco2_wang	TDC	regression	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_ rmse	0.342	ElasticNetCV	Conventional ML	22.979
tdc_clearance_ hepatocyte_az	TDC	regression	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_ rmse	43.841	CatBoost	Conventional ML	12.399
tdc_clearance_ microsome_az	TDC	regression	Uni-Mol (3D pretrained)	Uni-Mol V2 (84m)	test_ rmse	33.507	XGBoost	Conventional ML	16.477
tdc_half_life_ obach	TDC	regression	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_ rmse	18.541	Random forest	Conventional ML	21.086
tdc_ hydrationfreeenergy_ freesolv	TDC	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_ rmse	1.112	ElasticNetCV	Conventional ML	18.818
tdc_ld50_zhu	TDC	regression	Deep tabular NN (ChemML MLP)	ChemML MLP (Tensor Flow)	test_ rmse	0.828	XGBoost	Conventional ML	2.079
tdc_ lipophilicity_ astrazeneca	TDC	regression	Uni-Mol (3D pretrained)	Uni-Mol V1	test_ rmse	0.617	ChemML MLP (Py Torch)	Deep tabular NN (ChemML MLP)	17.583
tdc_ppbr_az	TDC	regression	MapLight + GNN	MapLight + GNN (Cat Boost, Strict Parity)	test_ rmse	11.356	ChemML MLP (Py Torch)	Deep tabular NN (ChemML MLP)	27.931
tdc_solubility_ aqsoldb	TDC	regression	Uni-Mol (3D pretrained)	Uni-Mol V1	test_ rmse	1.007	XGBoost	Conventional ML	1.814
tdc_vdss_ lombardo	TDC	regression	MapLight + GNN	MapLight + GNN (Cat Boost, Strict Parity)	test_ rmse	4.670	Extra trees	Conventional ML	50.704

Table S2: Staged component ablation: datasets whose best achievable primary metric improved when each pipeline stage was added, computed from the existing model results.

stage_order	stage	datasets_evaluated	datasets_improved_vs_previous	improvement_fraction
1	Conventional ML only	44	0	0.000
2	+ Map Light classic fea- tures	44	10	0.227
3	+ neu- ral/ deep back- ends	44	26	0.591
4	+ CFA fusion	44	6	0.136
5	Full pipeline incl. ensem- bles	44	16	0.364

Table S3: Feature-family selection summary across all datasets, sorted by per-feature enrichment relative to a uniform-selection baseline.

Feature family	Datasets selected	Available (sum)	Selected (sum)	% of selected	Enrichment vs uniform
rdkit	40	8,261	1,362	11.22	5.16
erg	38	12,552	1,146	9.44	2.86
avalon	43	44,384	3,603	29.69	2.54
maccs	38	5,962	224	1.85	1.18
maplight	41	30,068	854	7.04	0.89
fcfp6	43	42,500	998	8.22	0.73
atom_pair	41	39,651	830	6.84	0.66
rdk_path	40	45,036	896	7.38	0.62
topological_torsion	40	23,653	421	3.47	0.56
ecfp6	37	43,299	731	6.02	0.53
layered	40	43,642	674	5.55	0.48
morgan	38	40,827	398	3.28	0.31

Table S4: Model coverage: datasets attempted and datasets yielding a valid metric, per model.

model	datasets_attempted	datasets_valid
Uni-Mol V2 (164m)	4	0
Uni-Mol V2 (310m)	8	0
Chemprop v2 (CMPNN, ensemble=3)	44	6
Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)	44	6
Chemprop v2 (D-MPNN, ensemble=3)	44	6
Chemprop v2 (D-MPNN + Selected descriptors, ensemble=3)	44	6
Chemprop v2 (AttentiveFP, ensemble=3)	44	6
Uni-Mol V2 (84m)	44	17
ElasticNetCV	22	20
SVR	22	22
SVC	22	22
Voting Classifier (KNN, SVM)	22	22
MapLight CatBoost (Strict Parity)	22	22
Tabular CNN	22	22
LogisticRegression	22	22
Voting Regressor (KNN, SVM)	22	22
HistGradientBoosting	44	44
Extra trees	44	44
Ensemble (Weighted average (inverse train RMSE))	44	44
Ensemble (OOB Stacking (RidgeCV, 5-fold))	44	44
ChemML MLP (PyTorch)	44	44
CatBoost	44	44
CFA (Combinatorial Fusion)	44	44
ChemML MLP (TensorFlow)	44	44
AdaBoost	44	44
Tabular MLP	44	44
MapLight + GNN (CatBoost, Strict Parity)	44	44
Random forest	44	44
Uni-Mol V1	44	44
XGBoost	44	44

Table S5: Run-to-run comparison: the canonical NSF ACCESS Jetstream2 A100 benchmark against the earlier consumer-GPU (RTX 4060) run, analysed identically. “Same split” marks datasets where both runs used the identical held-out partition; the remainder were re-split to scaffold splits in the A100 run and are therefore not directly comparable. Change is relative and metric-direction aware (positive favours the A100 run).

Dataset	Task	Same split	A100 best model	A100 value	RTX 4060 best model	RTX 4060 value	Change (%)
tdc_ames	classification	True	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.876	XGBoost	0.875	0.142
tdc_bbb_martins	classification	True	Random forest	0.925	Random forest	0.932	-0.788
tdc_bioavailability_ma	classification	True	CatBoost	0.777	CatBoost	0.777	0.000
tdc_cyp1a2_veith	classification	True	Ensemble (Weighted average (inverse train RMSE))	0.971	Ensemble (Weighted average (inverse train RMSE))	0.970	0.091
tdc_cyp2c19_veith	classification	True	Ensemble (Weighted average (inverse train RMSE))	0.923	Ensemble (Weighted average (inverse train RMSE))	0.921	0.194
tdc_cyp2c9_substrate_carbonmangels	classification	True	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.466	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.482	-3.140
tdc_cyp2c9_veith	classification	True	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.814	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.810	0.481
tdc_cyp2d6_substrate_carbonmangels	classification	True	Uni-Mol V1	0.652	Ensemble (Weighted average (inverse train RMSE))	0.673	-3.096
tdc_cyp2d6_veith	classification	True	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.734	XGBoost	0.722	1.565
tdc_cyp3a4_substrate_carbonmangels	classification	True	Random forest	0.706	LogisticRegression	0.717	-1.494
tdc_cyp3a4_veith	classification	True	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.884	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.890	-0.637
tdc_dili	classification	True	Uni-Mol V1	0.922	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.915	0.760
tdc_herg	classification	True	AdaBoost	0.848	AdaBoost	0.861	-1.437
tdc_herg_karim	classification	True	Ensemble (Weighted average (inverse train RMSE))	0.905	Ensemble (Weighted average (inverse train RMSE))	0.902	0.355
tdc_hia_hou	classification	True	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.988	CFA (Combinatorial Fusion)	0.990	-0.187
tdc_pgp_broccatelli	classification	True	Uni-Mol V2 (84m)	0.933	Ensemble (Weighted average (inverse train RMSE))	0.929	0.423
chemml_cep_homo	regression	True	Ensemble (Weighted average (inverse train RMSE))	0.099	TabPFNRegressor	0.086	-15.648
chemml_organic_density	regression	True	Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)	0.005	TabPFNRegressor	0.005	-1.200
esol_delaney	regression	True	Ensemble (Weighted average (inverse train RMSE))	0.617	Ensemble (Weighted average (inverse train RMSE))	0.592	-4.220
freesolv_sampl	regression	True	ChemML MLP (PyTorch)	0.993	Chemprop v2 (AttentiveFP, ensemble=1)	1.080	8.015
lipophilicity	regression	True	Ensemble (Weighted average (inverse train RMSE))	0.589	Ensemble (Weighted average (inverse train RMSE))	0.582	-1.272

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Dataset	Task	Same split	A100 best model	A100 value	RTX 4060 best model	RTX 4060 value	Change (%)
poduam_pod_nc_std	regression	True	Random forest	0.720	Ensemble (Weighted average (inverse train RMSE))	0.699	-3.062
poduam_pod_rd_std	regression	True	Random forest	0.572	XGBoost	0.551	-3.896
polaris_adme_fang_hppb_1	regression	True	MapLight + GNN (CatBoost, Strict Parity)	0.448	MapLight + GNN (CatBoost, Strict Parity)	0.449	0.188
polaris_adme_fang_perm_1	regression	True	Uni-Mol V1	0.399	Uni-Mol V1	0.399	-0.040
polaris_adme_fang_rclint_1	regression	True	Uni-Mol V1	0.522	Uni-Mol V1	0.512	-1.976
polaris_adme_fang_rppb_1	regression	True	MapLight + GNN (CatBoost, Strict Parity)	0.493	MapLight + GNN (CatBoost, Strict Parity)	0.494	0.170
polaris_adme_fang_solu_1	regression	True	Uni-Mol V2 (84m)	0.551	Uni-Mol V1	0.575	4.202
tdc_caco2_wang	regression	True	Ensemble (Weighted average (inverse train RMSE))	0.342	Ensemble (Weighted average (inverse train RMSE))	0.336	-1.531
tdc_clearance_hepatocyte_az	regression	True	Ensemble (Weighted average (inverse train RMSE))	43.841	MapLight + GNN (CatBoost, Strict Parity)	43.976	0.308
tdc_clearance_microsome_az	regression	True	Uni-Mol V2 (84m)	33.507	Uni-Mol V1	35.888	6.634
tdc_half_life_obach	regression	True	Ensemble (Weighted average (inverse train RMSE))	18.541	Uni-Mol V1	17.222	-7.656
tdc_ld50_zhu	regression	True	ChemML MLP (TensorFlow)	0.828	CFA (Combinatorial Fusion)	0.834	0.687
tdc_lipophilicity_astrazeneca	regression	True	Uni-Mol V1	0.617	Uni-Mol V1	0.587	-5.135
tdc_ppbr_az	regression	True	MapLight + GNN (CatBoost, Strict Parity)	11.356	Ensemble (Weighted average (inverse train RMSE))	11.014	-3.108
tdc_solubility_aqsolddb	regression	True	Uni-Mol V1	1.007	Uni-Mol V1	0.989	-1.796
tdc_vdss_lombardo	regression	True	MapLight + GNN (CatBoost, Strict Parity)	4.670	MapLight + GNN (CatBoost, Strict Parity)	4.668	-0.035
tdc_carcinogens_lagunin	classification	False	Chemprop v2 (AttentiveFP, ensemble=3)	0.929	AdaBoost	0.863	7.656
tdc_clintox	classification	False	CFA (Combinatorial Fusion)	0.974	CatBoost	0.949	2.644
tdc_pampa_ncats	classification	False	Uni-Mol V1	0.763	Ensemble (Weighted average (inverse train RMSE))	0.806	-5.258
tdc_skin_reaction	classification	False	Uni-Mol V2 (84m)	0.658	CFA (Combinatorial Fusion)	0.769	-14.470
tdc_tox21	classification	False	AdaBoost	0.556	XGBoost	0.812	-31.520
tdc_toxcast	classification	False	HistGradientBoosting	0.716	CatBoost	0.791	-9.562
tdc_hydrationfreeenergy_freesolv	regression	False	Ensemble (OOF Stacking (RidgeCV, 5-fold))	1.112	CFA (Combinatorial Fusion)	0.556	-99.848

Table S6: Dataset catalog. “Ranking metric” is the metric used for cross-dataset model ranking (RMSE for regression; the dataset’s designated primary classification metric otherwise); leaderboard comparisons in Table S7 use the leaderboard’s own metric, which differs for several TDC regression tasks.

Dataset	Suite	Task	Molecules	Train	Test	Split	Target scale	Ranking metric	Best model	Best value	QSARena			Best published	Best published model
											Leaderboard metric	(lb metric)	Est. rank		
tdc_ames	TDC	classification	7,278	5,821	1,457	predefined	raw	roc_auc	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.876	ROC_-AUC	0.876	2	0.912	QW-MTL
tdc_bbb_martins	TDC	classification	2,030	1,624	406	predefined	raw	roc_auc	Random forest	0.925	ROC_-AUC	0.925	2	0.941	MolGPS (3B)
tdc_bioavailability_ma	TDC	classification	640	512	128	predefined	raw	roc_auc	CatBoost	0.777	ROC_-AUC	0.777	1	0.748	MaxQsaring
tdc_carcinogens_lagunin	TDC	classification	280	223	57	scaffold	raw	roc_auc	Chemprop v2 (AttentiveFP, ensemble=3)	0.929	ROC_-AUC	0.929	1	0.848	FATE-Tox (MTL)
tdc_clintox	TDC	classification	1,478	1,180	298	scaffold	raw	roc_auc	CFA (Combinatorial Fusion)	0.974	ROC_-AUC	0.974	3	0.996	PrismNet
tdc_cyp1a2_veith	TDC	classification	12,579	10,061	2,518	scaffold	raw	roc_auc	Ensemble (Weighted average (inverse train RMSE))	0.971					
tdc_cyp2c19_veith	TDC	classification	12,665	10,131	2,534	scaffold	raw	roc_auc	Ensemble (Weighted average (inverse train RMSE))	0.923					
tdc_cyp2c9_substrate_carbonmangels	TDC	classification	669	534	135	predefined	raw	auprc	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.466	AUPRC	0.466	1	0.450	MaxQsaring
tdc_cyp2c9_veith	TDC	classification	12,092	9,673	2,419	predefined	raw	auprc	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.814	AUPRC	0.814	3	0.877	MaxQsaring
tdc_cyp2d6_substrate_carbonmangels	TDC	classification	667	532	135	predefined	raw	auprc	Uni-Mol V1	0.652	AUPRC	0.652	5	0.766	MaxQsaring
tdc_cyp2d6_veith	TDC	classification	13,130	10,504	2,626	predefined	raw	auprc	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.734	AUPRC	0.734	2	0.811	MaxQsaring

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Dataset	Suite	Task	Molecules	Train	Test	Split	Target scale	Ranking metric	Best model	Best value	QSARena		Est. rank	Best published	Best published model
											Leaderboard metric	(lb metric)			
tdc_cyp3a4_substrate_carbonmangels	TDC	classification	670	535	135	predefined	raw	auprc	Random forest	0.706	ROC_AUC	0.655	7	0.692	MolE
tdc_cyp3a4_veith	TDC	classification	12,328	9,861	2,467	predefined	raw	auprc	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.884	AUPRC	0.884	2	0.923	MaxQsaring
tdc_dili	TDC	classification	475	379	96	predefined	raw	roc_auc	Uni-Mol V1	0.922	ROC_AUC	0.922	4	0.945	Meta-model (NIST)
tdc_herg	TDC	classification	655	523	132	predefined	raw	roc_auc	AdaBoost	0.848	ROC_AUC	0.848	4	0.880	MaxQsaring
tdc_herg_karim	TDC	classification	13,445	10,755	2,690	scaffold	raw	roc_auc	Ensemble (Weighted average (inverse train RMSE))	0.905					
tdc_hia_hou	TDC	classification	578	461	117	predefined	raw	roc_auc	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.988	ROC_AUC	0.988	4	0.994	MiniMol (GINE)
tdc_pampa_ncats	TDC	classification	2,034	1,626	408	scaffold	raw	roc_auc	Uni-Mol V1	0.763					
tdc_pgp_broccatelli	TDC	classification	1,218	973	245	predefined	raw	roc_auc	Uni-Mol V2 (84m)	0.933	ROC_AUC	0.933	4	0.994	MiniMol (GINE)
tdc_skin_reaction	TDC	classification	404	289	115	scaffold	raw	roc_auc	Uni-Mol V2 (84m)	0.658	ROC_AUC	0.658	>10	0.741	FATE-Tox (MTL)
tdc_tox21	TDC	classification	7,258	5,797	1,461	scaffold	raw	roc_auc	AdaBoost	0.556	ROC_AUC	0.556	>10	0.867	PrismNet
tdc_toxcast	TDC	classification	1,731	1,357	374	scaffold	raw	roc_auc	HistGradient Boosting	0.716	ROC_AUC	0.716	1	0.714	PrismNet
chemml_cep_homo	ChemML	regression	500	400	100	target_	raw	rmse	Ensemble (Weighted average (inverse train RMSE))	0.099					
chemml_organic_density	ChemML	regression	500	400	100	target_	log10	rmse	Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)	0.005					
esol_delaney	MoleculeNet	regression	1,128	874	254	scaffold	raw	rmse	Ensemble (Weighted average (inverse train RMSE))	0.617	RMSE	0.617	6	0.558	GCN
freesolv_sampl	MoleculeNet	regression	642	513	129	random	raw	rmse	ChemML MLP (Py Torch)	0.993					

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Dataset	Suite	Task	Molecules	Train	Test	Split	Target scale	Ranking metric	Best model	Best value	Leaderboard metric	QSARena (lb metric)	Est. rank	Best published	Best published model
lipophilicity	MoleculeNet	regression	4,200	3,357	843	scaffold	raw	rmse	Ensemble (Weighted average (inverse train RMSE))	0.589	RMSE	0.589	7	0.549	GCN
poduam_pod_nc_std	PODUAM	regression	1,842	1,473	369	target_- raw	raw	rmse	Random forest	0.720	RMSE	0.720	2	0.550	PODUAM BNN (PODnc)
poduam_pod_rd_std	PODUAM	regression	2,355	1,884	471	quartiles target_- raw	raw	rmse	Random forest	0.572	RMSE	0.572	2	0.410	PODUAM BNN (PODrd)
polaris_adme_fang_hppb_1	Polaris	regression	1,808	1,446	362	quartiles predefined	raw	rmse	MapLight + GNN (CatBoost, Strict Parity)	0.448	MSE	0.201	3	0.143	1B_MPNN_LargeMix-and-Phenomics
polaris_adme_fang_perm_1	Polaris	regression	2,642	2,113	529	predefined	raw	rmse	Uni-Mol V1	0.399	MSE	0.159	3	0.113	1B_MPNN_MolGPS-ens_LargeMix
polaris_adme_fang_rclint_1	Polaris	regression	3,054	2,443	611	predefined	raw	rmse	Uni-Mol V1	0.522	MSE	0.273	4	0.216	1B_MPNN_MolGPS-ens_LargeMix
polaris_adme_fang_rppb_1	Polaris	regression	885	708	177	predefined	raw	rmse	MapLight + GNN (CatBoost, Strict Parity)	0.493	MSE	0.243	3	0.230	1B_MPNN_LargeMix-and-Phenomics
polaris_adme_fang_solu_1	Polaris	regression	2,173	1,738	435	predefined	raw	rmse	Uni-Mol V2 (84m)	0.551	MSE	0.304	5	0.222	1B_MPNN_MolGPS-ens_LargeMix
tdc_caco2_wang	TDC	regression	910	728	182	predefined	raw	rmse	Ensemble (Weighted average (inverse train RMSE))	0.342	MAE	0.272	3	0.256	CaliciBoost
tdc_clearance_hepatocyte_az	TDC	regression	1,213	970	243	predefined	raw	rmse	Ensemble (Weighted average (inverse train RMSE))	43.841	SPEARMAN	0.516	3	0.633	CFA

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Dataset	Suite	Task	Molecules	Train	Test	Split	Target scale	Ranking metric	Best model	Best value	Leaderboard metric	QSARena (lb metric)	Est. rank	Best published	Best published model
tdc_clearance_microsome_az	TDC	regression	1,102	881	221	predefined	raw	rmse	Uni-Mol V2 (84m)	33.507	SPEARMAN	0.678	1	0.652	MapLight + GNN
tdc_half_life_obach	TDC	regression	667	532	135	predefined	raw	rmse	Ensemble (Weighted average (inverse train RMSE))	18.541	SPEARMAN	0.597	2	0.649	CFA
tdc_hydrationfreeenergy_freesolv	TDC	regression	642	490	152	scaffold	raw	rmse	Ensemble (OOF Stacking (RidgeCV, 5-fold))	1.112	RMSE	1.112	9	0.654	PrismNet
tdc_ld50_zhu	TDC	regression	7,385	5,907	1,478	predefined	raw	rmse	ChemML MLP (TensorFlow)	0.828	MAE	0.575	5	0.292	Base Boosting KyQVZ6b2
tdc_lipophilicity_astrazeneca	TDC	regression	4,200	3,360	840	predefined	raw	rmse	Uni-Mol V1	0.617	MAE	0.470	7	0.406	MiniMol
tdc_ppbr_az	TDC	regression	2,790	2,231	559	predefined	raw	rmse	MapLight + GNN (CatBoost, Strict Parity)	11.356	MAE	7.306	3	0.679	Gradient Boost
tdc_solubility_aqsoldb	TDC	regression	9,980	7,985	1,995	predefined	raw	rmse	Uni-Mol V1	1.007	MAE	0.710	2	0.557	MiniMol
tdc_vdss_lombardo	TDC	regression	1,130	904	226	predefined	raw	rmse	MapLight + GNN (CatBoost, Strict Parity)	4.670	SPEARMAN	0.680	5	0.942	MapLight + GNN

Table S7: Per-dataset leaderboard comparison, sorted by estimated rank. “References (n)” is the number of published values available for that dataset and metric; ranks from sparse reference sets are correspondingly uncertain. The **tdc_ppbr_az** top-1 reference (MAE 0.679) is inconsistent in scale with the rest of that dataset’s references (top-10 cutoff 7.914) and its top-1 gap should be disregarded.

Dataset	Metric	QSARena best model (test-selected)	QSARena value	Reference top-1	Reference top-10 cutoff	References (n)	Est. rank	CV-selected model	CV-selected value	CV-selected est. rank
tdc_bioavailability_ma	ROC_	CatBoost	0.777	0.748	0.640	8	1	Logistic Regression	0.715	3
tdc_carcinogens_lagunin	AUC									
	ROC_	Chemprop v2 (AttentiveFP, ensemble=3)	0.929	0.848	0.795	20	1	Voting Classifier (KNN, SVM)	0.837	5
tdc_clearance_microsome_az	AUC									
	SPEARMAN	UniMol V2 (84m)	0.678	0.652	0.599	17	1	XGBoost	0.305	>10
tdc_cyp2c9_substrate_carbonmangels	AUPRC	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.466	0.450	0.360	6	1	SVC	0.301	7
tdc_toxcast	ROC_	HistGradientBoosting	0.716	0.714	0.714	2	1	Logistic Regression	0.595	3
poduam_pod_nc_std	AUC									
	RMSE	Random forest	0.720	0.550	0.730	2	2	ElasticNetCV	0.792	3
poduam_pod_rd_std	RMSE	Random forest	0.572	0.410	0.630	2	2	ElasticNetCV	0.703	3
tdc_ames	ROC_	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.876	0.912	0.834	7	2	Extra trees	0.871	2
tdc_bbb_martins	AUC									
	ROC_	Random forest	0.925	0.941	0.903	7	2	Tabular MLP	0.898	8
tdc_cyp2d6_veith	AUC									
	AUPRC	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.734	0.811	0.464	6	2	AdaBoost	0.615	6
tdc_cyp3a4_veith	AUPRC	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.884	0.923	0.750	6	2	AdaBoost	0.798	6
tdc_half_life_obach	SPEARMAN	UniMol V1	0.597	0.649	0.485	16	2	Random forest	0.108	>10
tdc_solubility_aqsolddb	MAE	Uni-Mol V1	0.710	0.557	0.776	17	2	XGBoost	0.744	6
polaris_adme_fang_hppb_1	MSE	MapLight + GNN (Cat Boost, Strict Parity)	0.201	0.143	0.383	10	3	ElasticNetCV	0.300	8
polaris_adme_fang_perm_1	MSE	Uni-Mol V1	0.159	0.113	0.257	10	3	ChemML MLP (PyTorch)	0.212	7

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Dataset	Metric	QSARena best model (test-selected)	QSARena value	Reference top-1	Reference top-10 cutoff	References (n)	Est. rank	CV-selected model	CV-selected value	CV- selected est. rank
polaris_adme_ fang_rppb_1	MSE	MapLight + GNN (Cat Boost, Strict Parity)	0.243	0.230	0.634	10	3	ElasticNetCV	0.361	5
tdc_caco2_wang	MAE	MapLight CatBoost (Strict Parity)	0.272	0.256	0.288	20	3	ElasticNetCV	0.345	>10
tdc_clearance_ hepatocyte_az	SPEARMAP	MapLight + GNN (Cat Boost, Strict Parity)	0.516	0.633	0.440	16	3	CatBoost	0.224	>10
tdc_clintox	ROC_ - AUC	CFA (Combinatorial Fusion)	0.974	0.996	0.889	12	3	Logistic Regression	0.896	5
tdc_cyp2c9_veith	AUPRC	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.814	0.877	0.770	6	3	AdaBoost	0.705	7
tdc_ppbr_az	MAE	MapLight + GNN (Cat Boost, Strict Parity)	7.306	0.679	7.914	16	3	ChemML MLP (PyTorch)	9.741	>10
polaris_adme_ fang_rclint_1	MSE	Uni-Mol V1	0.273	0.216	0.403	10	4	ElasticNetCV	0.365	10
tdc_dili	ROC_ - AUC	Uni-Mol V1	0.922	0.945	0.852	6	4	Logistic Regression	0.745	7
tdc_herg	ROC_ - AUC	AdaBoost	0.848	0.880	0.806	7	4	Logistic Regression	0.709	8
tdc_hia_hou	ROC_ - AUC	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.988	0.994	0.976	8	4	Logistic Regression	0.983	7
tdc_pgp_ broccatelli	ROC_ - AUC	Uni-Mol V2 (84m)	0.933	0.994	0.911	8	4	Logistic Regression	0.868	9
polaris_adme_ fang_solu_1	MSE	Uni-Mol V2 (84m)	0.304	0.222	0.329	10	5	ChemML MLP (PyTorch)	0.410	>10
tdc_cyp2d6_ substrate_ carbonmangels	AUPRC	Uni-Mol V1	0.652	0.766	0.570	7	5	AdaBoost	0.620	7
tdc_ld50_zhu	MAE	Ensemble (Weighted average (inverse train RMSE))	0.575	0.292	0.605	16	5	XGBoost	0.588	8
tdc_vdss_ lombardo	SPEARMAP	MapLight + GNN (Cat Boost, Strict Parity)	0.680	0.942	0.582	16	5	Extra trees	0.368	>10
esol_delaney	RMSE	Ensemble (Weighted average (inverse train RMSE))	0.617	0.558	0.743	30	6	ElasticNetCV	0.726	9
lipophilicity	RMSE	Ensemble (Weighted average (inverse train RMSE))	0.589	0.549	0.610	27	7	ElasticNetCV	0.682	>10

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Dataset	Metric	QSARena best model (test-selected)	QSARena value	Reference top-1	Reference top-10 cutoff	References (n)	Est. rank	CV-selected model	CV-selected value	CV- selected est. rank
tdc_cyp3a4_ substrate_ carbonmangels	ROC_ - AUC	Voting Classifier (KNN, SVM)	0.655	0.692	0.651	7	7	Logistic Regression	0.624	8
tdc_lipophilicity_ astrazeneca	MAE	CFA (Combinatorial Fusion)	0.470	0.406	0.515	17	7	ChemML MLP (PyTorch)	0.542	>10
tdc_ hydrationfreeenergy_ freesolv	RMSE	Ensemble (OOF Stacking (RidgeCV, 5-fold))	1.112	0.654	1.211	12	9	ElasticNetCV	1.321	>10
tdc_skin_reaction	ROC_ - AUC	Uni-Mol V2 (84m)	0.658	0.741	0.677	21	>10	Voting Classifier (KNN, SVM)	0.503	>10
tdc_tox21	ROC_ - AUC	AdaBoost	0.556	0.867	0.840	12	>10	SVC	0.467	>10

Table S8: Applicability domain and calibration on the 22 official TDC ADMET Benchmark Group splits, for a fixed reference model (random forest, 300 trees, on Morgan radius-2 2048-bit fingerprints plus RDKit 2D descriptors) fitted to 80% of each train_val partition, with the remaining 20% used only for conformal calibration. Coverage is the fraction of test compounds inside the domain. The error ratio is the mean test error outside the domain divided by that inside it (MAE for regression, 0/1 error for classification); values above 1 mean the domain flag separates worse predictions. “Std.” is the Roy standardization approach on RDKit descriptors; “kNN” is the 5-nearest-neighbour Tanimoto distance thresholded at the 95th percentile of training distances; “Reliability” is the conformal-interval-width flag (regression) or the posterior-probability confidence flag with $|p_1 - p_0| \geq 0.5$ (classification). Conformal coverage is the empirical test coverage of 90% split-conformal intervals or prediction sets. “—” marks a ratio not reported because fewer than five test compounds fall on one side of the domain boundary. Results describe the reference model, not the benchmark’s selected model; single seed.

Dataset	Task	n test	Std. coverage	Std. ratio	kNN coverage	kNN ratio	Reliability ratio	Conformal coverage	ECE
caco2_wang	reg.	182	0.99	—	0.74	0.75	1.38	0.95	—
clearance_hepatocyte_az	reg.	243	0.94	1.09	0.88	1.26	1.75	0.94	—
clearance_microsome_az	reg.	221	0.99	—	0.97	1.32	2.08	0.93	—
half_life_obach	reg.	135	0.95	1.36	0.94	0.87	3.60	0.94	—
ld50_zhu	reg.	1478	0.95	1.16	0.82	1.24	2.11	0.82	—
lipophilicity_astazeneca	reg.	840	0.99	2.33	0.94	1.55	1.47	0.88	—
ppbr_az	reg.	559	0.99	1.35	0.91	1.05	3.01	0.93	—
solubility_aqsolddb	reg.	1995	0.95	1.45	0.86	1.15	1.59	0.89	—
vdss_lombardo	reg.	226	0.94	0.49	0.86	1.22	8.26	0.95	—
ames	cls.	1457	0.96	0.91	0.84	1.71	2.67	0.89	0.07
bbb_martins	cls.	406	0.96	0.00	0.78	1.19	9.30	0.91	0.06
bioavailability_ma	cls.	128	0.98	—	0.96	0.72	3.72	0.91	0.08
cyp2c9_substrate_carbonmangels	cls.	135	0.93	0.68	0.95	0.00	2.15	0.81	0.11
cyp2c9_veith	cls.	2419	0.96	0.53	0.88	0.62	5.21	0.91	0.05
cyp2d6_substrate_carbonmangels	cls.	135	0.96	0.65	0.98	—	4.82	0.89	0.06
cyp2d6_veith	cls.	2626	0.96	0.90	0.84	0.89	5.29	0.91	0.06
cyp3a4_substrate_carbonmangels	cls.	135	0.95	2.08	0.94	1.80	1.47	0.93	0.06
cyp3a4_veith	cls.	2467	0.98	0.91	0.94	0.93	4.88	0.89	0.07
dili	cls.	96	0.95	3.64	0.95	0.00	2.16	0.95	0.17
herg	cls.	132	0.99	—	0.92	1.10	4.71	0.86	0.10
hia_hou	cls.	117	0.97	—	0.82	4.57	6.13	0.91	0.09
pgp_broccatelli	cls.	245	0.94	0.75	0.84	1.21	5.04	0.88	0.06